

PART I

Output

```
C:\Users\ADMIN\Desktop\Codes>cd "c
1. true, 10, false, 15
2. true, false, 25, 30
3. 120, 70, 144, 16
```

Code

```
1  class NumOne {
2      int a = 9;
3      int b = 6;
4      int c = 42;
5      boolean d = true;
6      boolean e = false;
7
8      boolean resultNumOne_1 = d | e;           // logical bitwise OR (|)
9      int resultNumOne_2 = c >> 2;             // right shift (>>) by 2
10     boolean resultNumOne_3 = d & e;           // logical bitwise AND (&)
11     int resultNumOne_4 = a | b;              // logical bitwise OR (|)
12
13     public void displayNumOne() {
14         System.out.println("1. " + resultNumOne_1 + ", " + resultNumOne_2 + ", " + resultNumOne_3 + ", " + resultNumOne_4);
15     }
16 }
17
18 class NumTwo {
19     int a = 25;
20     int b = 30;
21     int c = 50;
22
23     String resultNumTwo_1 = (b > a) ? "true" : "false"; // if b is greater than a, return true, else return false
24     // **turned from boolean to String to avoid method redundancy
25     String resultNumTwo_2 = (b < a) ? "true" : "false"; // if b is less than a, return true, else return false
26     // **turned from boolean to String to avoid method redundancy
27     int resultNumTwo_3 = (a < c) ? a : c; // if a is less than c, return a, else return c (min)
28     int resultNumTwo_4 = (a == b) ? a : b; // is b is equal to c, return c, else return b
29
30     public void displayNumTwo() {
31         System.out.println("2. " + resultNumTwo_1 + ", " + resultNumTwo_2 + ", " + resultNumTwo_3 + ", " + resultNumTwo_4);
32     }
33 }
34
35 class NumThree {
36     int a = 60;
37     int b = 70;
38     int c = 12;
39     int d = 16;
40     int e = 140;
41     int f = 256;
42
43     int resultNumThree_1 = a;
44     int resultNumThree_2 = e;
45     int resultNumThree_3 = c;
46     int resultNumThree_4 = f;
47
48     public void computeNumThree() {
49         resultNumThree_1 += a; // addition assignment operator (+=)
50         resultNumThree_2 -= b; // subtraction assignment operator (-=)
51         resultNumThree_3 *= c; // multiplication assignment operator (*=)
52         resultNumThree_4 /= d; // division assignment operator (/=)
53     }
54
55     public void displayNumThree() {
56         System.out.println("3. " + resultNumThree_1 + ", " + resultNumThree_2 + ", " + resultNumThree_3 + ", " + resultNumThree_4);
57     }
58 }
59
```

```

60 public class ActOne_PartOne {
    Run main | Debug main | Run | Debug
61     public static void main(String[] args) {
62         NumOne obj1 = new NumOne();
63         NumTwo obj2 = new NumTwo();
64         NumThree obj3 = new NumThree();
65
66         obj1.displayNumOne();
67         obj2.displayNumTwo();
68         obj3.computeNumThree();
69         obj3.displayNumThree();
70     }
71 }

```

PART II

Output

```

C:\Users\ADMIN\Desktop\Codes>cd "c:\Users\ADMIN\Desktop\Codes\Java\"
--- Student Info ---
[Student 1]
Name: Seele
Math Grade: 89.43
Science Grade: 93.68

GWA: 91.56
Passed? true
Performance: Excellent

[Student 2]
Name: Bronya
Math Grade: 99.35
Science Grade: 97.89

GWA: 98.62
Passed? true
Performance: Outstanding

[Student 3]
Name: Hook
Math Grade: 73.56
Science Grade: 71.67

GWA: 72.61
Passed? false
Performance: Failed

--- Professor Ratings ---
Math Professor: 3.23
Satisfactory Performance? false

Science Professor: 3.67
Satisfactory Performance? true

```

Code

```
1 // Creates 'Student' object
2 class Student {
3     String name;
4     float gradeMath;
5     float gradeScience;
6     float profMathRating;
7     float profScienceRating;
8
9     public Student(String n, float m, float s, float mr, float sr) {
10         name = n;
11         gradeMath = m;
12         gradeScience = s;
13         profMathRating = mr;
14         profScienceRating = sr;
15     }
16
17     // Computes GWA of the student
18     public float computeGWA() {
19         return (gradeMath + gradeScience) / 2;
20     }
21
22     // Determines if the student passed based on GWA
23     public boolean isPassed() {
24         return computeGWA() >= 75;
25     }
26
27     // Determines the performance of the student
28     public String getPerformance() {
29         float gwa = computeGWA();
30
31         if (gwa >= 75 && gwa < 85)
32             return "Satisfactory";
33         else if (gwa >= 85 && gwa <= 90)
34             return "Very Satisfactory";
35         else if (gwa > 90 && gwa < 95)
36             return "Excellent";
37         else if (gwa >= 95 && gwa <= 100)
38             return "Outstanding";
39         else
40             return "Failed";
41     }
42 }
43
44 // Main class
45 public class ActOne_PartTwo {
46
47     Run main | Debug main | Run | Debug
48     public static void main(String[] args) {
49         // Creates an array of Students
50         Student[] s = {
51             new Student(n: "Seele", m: 89.43f, s: 93.68f, mr: 3, sr: 3),
52             new Student(n: "Bronya", m: 99.35f, s: 97.89f, mr: 4.7f, sr: 5),
53             new Student(n: "Hook", m: 73.56f, s: 71.67f, mr: 2, sr: 3)
54         };
55
56         // Prints the array of Students and their info
57         System.out.println(x: "--- Student Info ---");
58
59         for (int i = 0; i < s.length; i++) {
60             System.out.println("[Student " + (i + 1) + "]");
61             System.out.println("Name: " + s[i].name);
62             System.out.printf(format: "Math Grade: %.2f\n", s[i].gradeMath);
63             System.out.printf(format: "Science Grade: %.2f\n", s[i].gradeScience);
64             System.out.println();
65             // Prints their average and determines their performance
66             System.out.printf(format: "GWA: %.2f\n", s[i].computeGWA()); // Numeric output
67             System.out.println("Passed? " + s[i].isPassed()); // Boolean output
68             System.out.println("Performance: " + s[i].getPerformance());
69             System.out.println();
70         }
71
72         // Calculates Professor ratings
73         float totalMath = 0;
74         float totalScience = 0;
```

← nalogay nung iniiscreenshot ko na po, pero wala yan nung ni-run ko hehe T_T

```
75     for (int i = 0; i < s.length; i++) {
76         totalMath += s[i].profMathRating;
77         totalScience += s[i].profScienceRating;
78     }
79
80     float avgMath = totalMath / s.length;
81     float avgScience = totalScience / s.length;
82
83     boolean mathProfPerformance = avgMath >= 3.5f;
84     boolean scienceProfPerformance = avgScience >= 3.5f;
85
86     // Prints Professor ratings
87     System.out.println(x: "--- Professor Ratings ---");
88     System.out.printf(format: "Math Professor: %.2f\n", avgMath);           // Numeric output
89     System.out.println("Satisfactory Performance? " + mathProfPerformance); // Boolean output
90     System.out.println();
91
92     System.out.printf(format: "Science Professor: %.2f\n", avgScience);     // Numeric output
93     System.out.println("Satisfactory Performance? " + scienceProfPerformance); // Boolean output
94 }
95 }
96
```